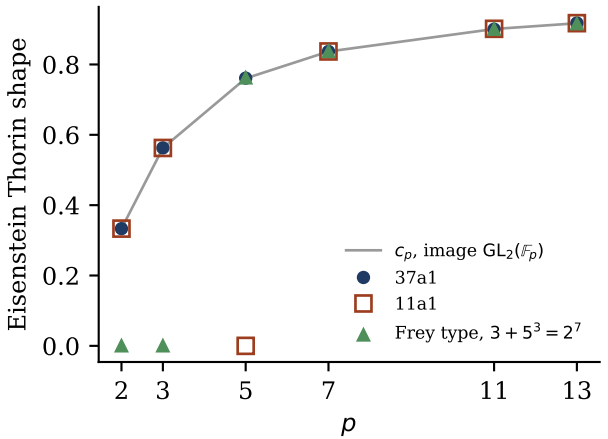
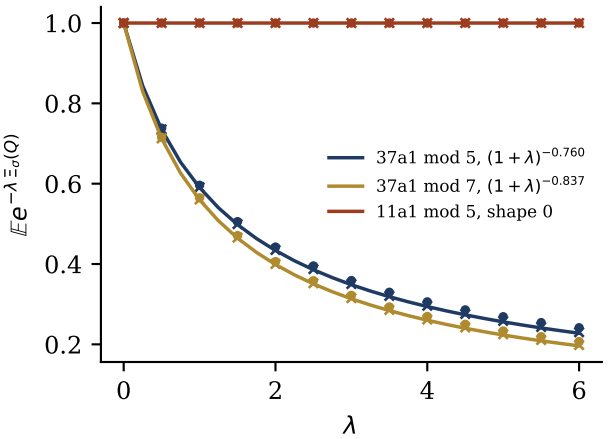

shape $c(E,p)$: zero means Eisenstein mod p
Eisenstein Thorin shape
0.0
0.2
0.4
0.6
0.8
2
3
5
7
11
13
p
c_p, image $\mathrm{GL}_2(\mathbb{F}_p)$
37a1
11a1
Frey type, $3+5^3=2^7$


Gamma limits; dots $\sigma=1.01$, crosses $\sigma=1.002$
$\mathbb{E}\,e^{-\lambda\,\Xi_\sigma(Q)}$
1.0
0.8
0.6
0.4
0.2
0
2
4
6
λ
37a1 mod 5, $(1+\lambda)^{-0.760}$
37a1 mod 7, $(1+\lambda)^{-0.837}$
11a1 mod 5, shape 0